Machine Learning for Physics

Research projects

Master of Science in Industrial and Applied Mathematics, 2026/2027

Project assignment

Work individually or in pairs.
Select one of the papers below and investigate the methods and experiments in two successive stages: first reproduce selected results from the paper, then propose and test a simple improvement motivated by your findings. The purpose is to understand the method, assess its limitations, and draw conclusions supported by numerical evidence.

The project should have two connected parts:

1. Reproduce part or all of the experiments in the chosen paper. Start from results that you consider important for understanding or assessing the method. You are not required to reproduce all the experiments: some papers involve datasets, training budgets, or numerical experiments that may be too large. Define the scope of your project yourselves and justify your choices, considering scientific relevance, computational cost, available resources, and what can be concluded from the selected experiments (below, some suggestions are given to guide you). Explain differences, if any, from the article's workflow/method and discuss agreements or discrepancies with the original results.

2. Propose a motivated improvement to the approach and validate it with at least one numerical experiment. Use the reproduction as a baseline and identify a limitation, assumption, or design choice worth revisiting. The change could concern the model, physical constraints, data, training, or numerical formulation. Evaluate whether it helps, using suitable comparisons and diagnostics. An improvement need not succeed: a carefully tested negative result, with an explanation of what was learned, is also valuable.

Discussion should accompany both parts. The suggestions below are starting points. You are responsible for choosing a coherent, feasible project and set of experiment(s) to reproduce and for explaining your choices. The assessment will reflect the ambition and relevance of the work, the quality of its implementation and validation, and the depth of the analysis.

Note: a project limited to a minimal reproduction with little analysis, an untested improvement proposal, or simply running existing code without understanding it will receive a correspondingly low mark. The number of experiments alone does not determine the quality of the work.

Written report, code and oral examination

Submit a scientific report and the code needed to reproduce the aforementioned results by 16 January 2027 via email. The report should distinguish the two parts of the project:

1. Which experiments you selected, why they represent an appropriate scope, how you reproduced them, and how your results compare with those reported in the paper,

2. What improvement you proposed, why you proposed it, and whether the comparison with your reproduced baseline supports its effectiveness.

The code should be reproducible and usable by others. Include instructions for running it, dependencies, data sources, relevant parameters and random seeds, and information on how to regenerate the main results. You may use publicly available implementations and existing libraries, provided that you acknowledge them and clearly explain your own contribution.

You will also present and defend your project orally between 18 and 22 January 2027. Each student, including both members of a pair (if any), must be able to explain the project individually.

List of papers

1. Discovering governing equations from data: Sparse identification of nonlinear dynamical systems

Authors: S. L. Brunton, J. L. Proctor, and J. N. Kutz
Year: 2015
URL: https://arxiv.org/pdf/1509.03580.
Suggested experiments: Sections 4.1–4.2.

2. Hamiltonian neural networks

Authors: S. Greydanus, M. Dzamba, and J. Yosinski
Year: 2019
URL: https://arxiv.org/pdf/1906.01563.
Suggested experiments: Section 3 (Section 4 as an extension).

3. Symmetry-aware autoencoders: s-PCA and s-NLPCA

Authors: S. Kneer, T. Sayadi, D. Sipp, P. Schmid, and G. Rigasa
Year: 2021
URL: https://arxiv.org/pdf/2111.02893
Suggested experiments: Section 4, with emphasis on discrete symmetries.

4. Stabilized neural ordinary differential equations for long-time forecasting of dynamical systems

Authors: A. J. Linot, J. W. Burby, Q. Tang, P. Balaprakash, M. D. Graham, and R. Maulik
Year: 2022
URL: https://arxiv.org/pdf/2203.15706v2.

5. Physics-informed neural operator for learning partial differential equations

Authors: Z. Li, H. Zheng, N. Kovachki, D. Jin, H. Chen, B. Liu, K. Azizzadenesheli, and A. Anandkumar
Year: 2023
URL: https://arxiv.org/abs/2111.03794.
Suggested experiments: Section 4.1, Table 1 (Burgers' equation and Darcy flow).

6. PINNs failure modes are overfitting

Authors: N. T. Andersen and T. Matsubara
Year: 2026
URL: https://arxiv.org/pdf/2605.30910.
Suggestion: Section 4.1 (with Sections 4.2 and 5 as possible follow-ups).